Methodology

Pipeline

Ground truth = Dorado SUP basecaller → minimap2 (RawBench protocol).
RawHash2 invoked with -x sensitive (and --r10 for R10.4.x), Uncalled4 pore model.
Eval = best-mapping per read, 0.1 overlap threshold (TP/FP/FN → F1).
Read-ID-mismatch fix on D3/D4: re-extract reads from FAST5 internal fastq → minimap2.

Segmenters (9, 5 algorithmic families)

- t-test family: default (ONT dual-window), scrappie
- DP-CPD: pelt (BIC penalty), binseg (per-segment penalty fix)
- Probabilistic: hmm (Viterbi+k-means), bocd (Adams & MacKay 2007)
- Sliding-window: window (z-score), mad (median two-sample)
- Edge: gradient (1st derivative)

Top-K mechanism (key engineering)

Replaces threshold-based detection with rank-based selection: pick the K=n/9 highest-scoring positions (where 9 = R10 samples-per-k-mer). Greedy NMS with min_size=4 separation. K is derived from pore physics, not from data → not overfit. Boost on gradient/mad/window: F1 ≈0 → 0.1–0.5 (RB_ecoli).

Per-segmenter index

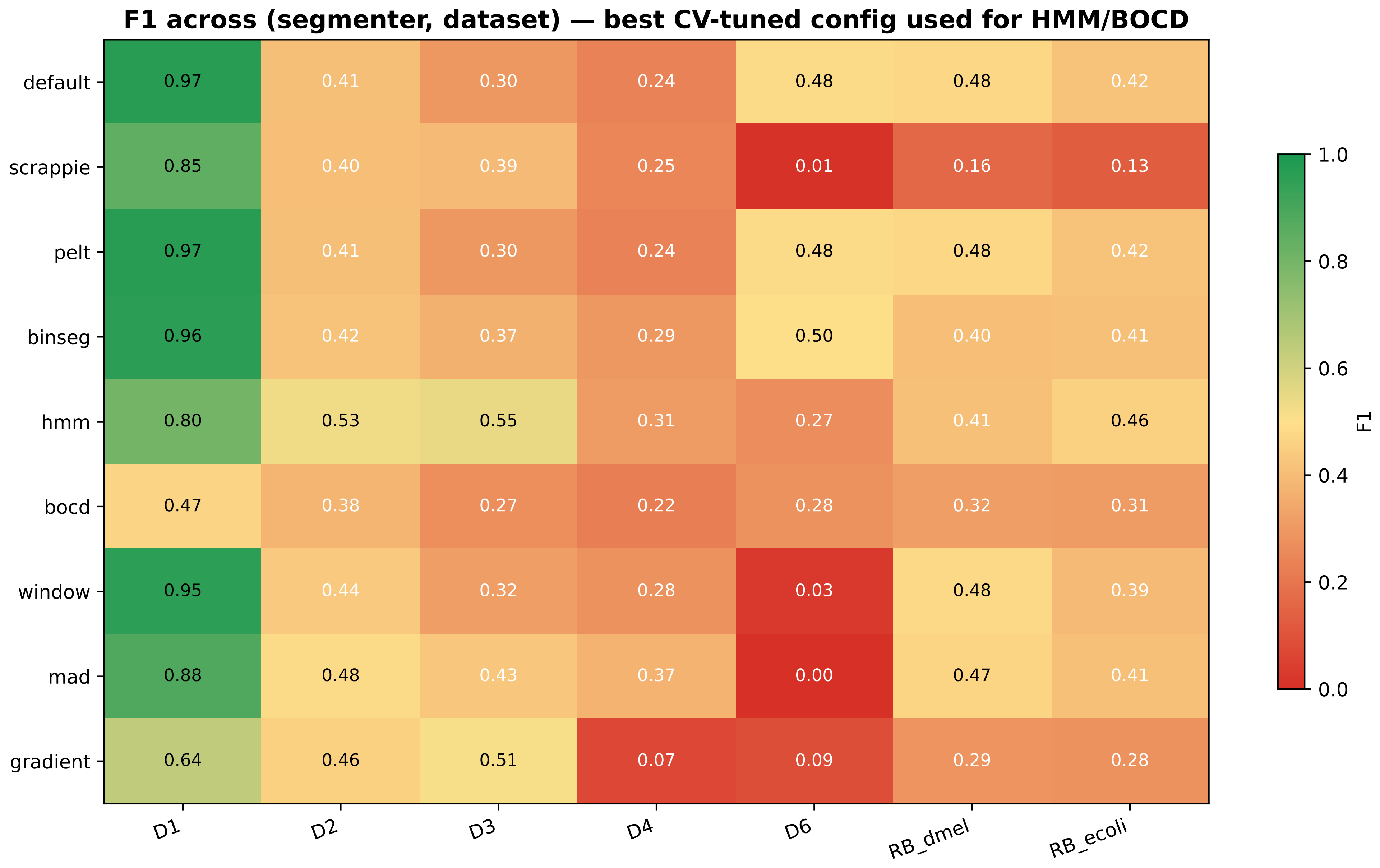
Each segmenter builds its OWN RawHash2 index by simulating reference signal + segmenting it with the same method used at query time. Without per-segmenter index, non-default segmenters give F1≈0 due to feature-distribution mismatch.

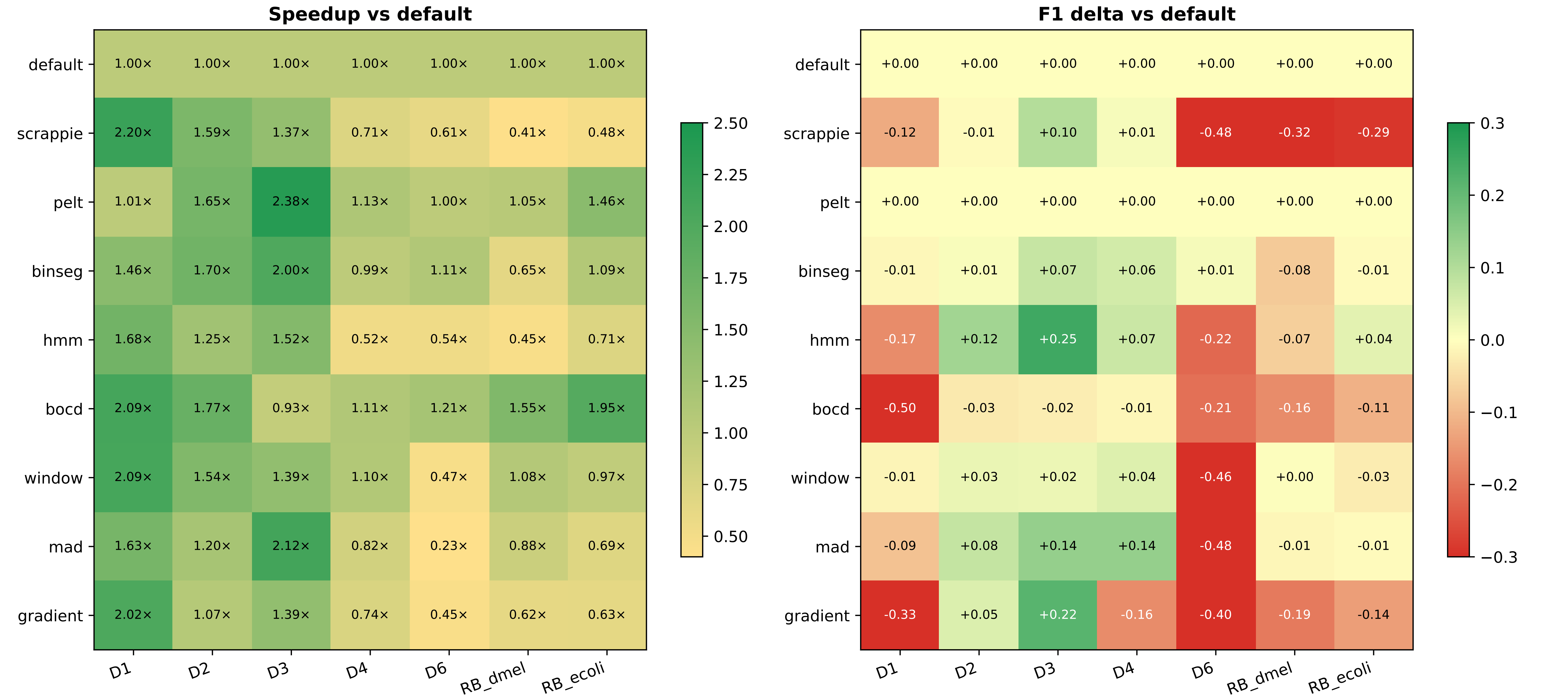
Cross-validation

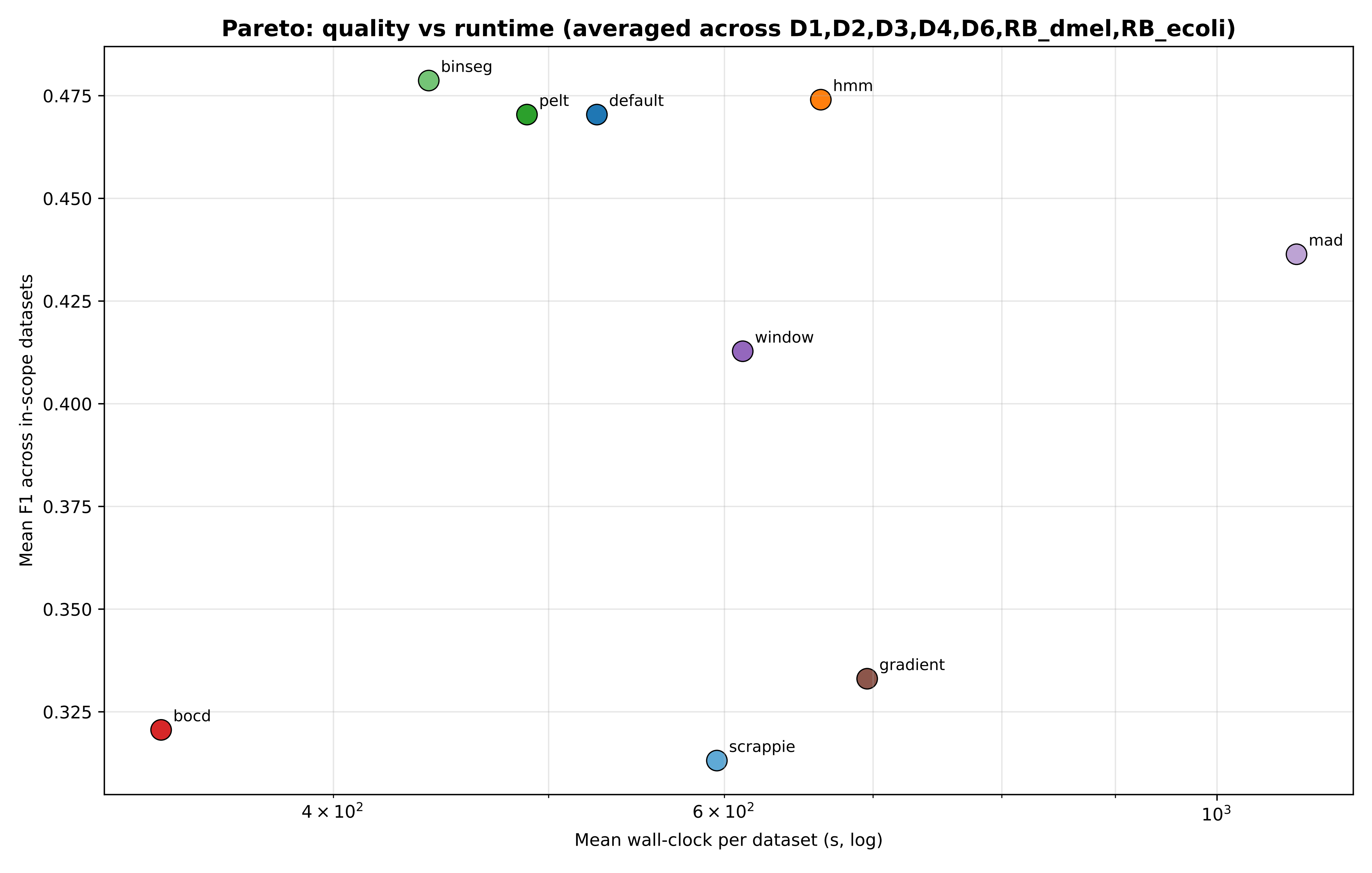
3-fold read-level CV on D1/D2/D4/D6/RB_ecoli/RB_dmel for HMM (12 configs: STATES ∈ {3..8} × STAY ∈ {0.93, 0.96}) and BOCD (6 configs: EXP_LEN ∈ {50..600}). Train-test gap nearly 0 across all configs → no overfit. Best per dataset selected by mean F1_test (not F1_train).

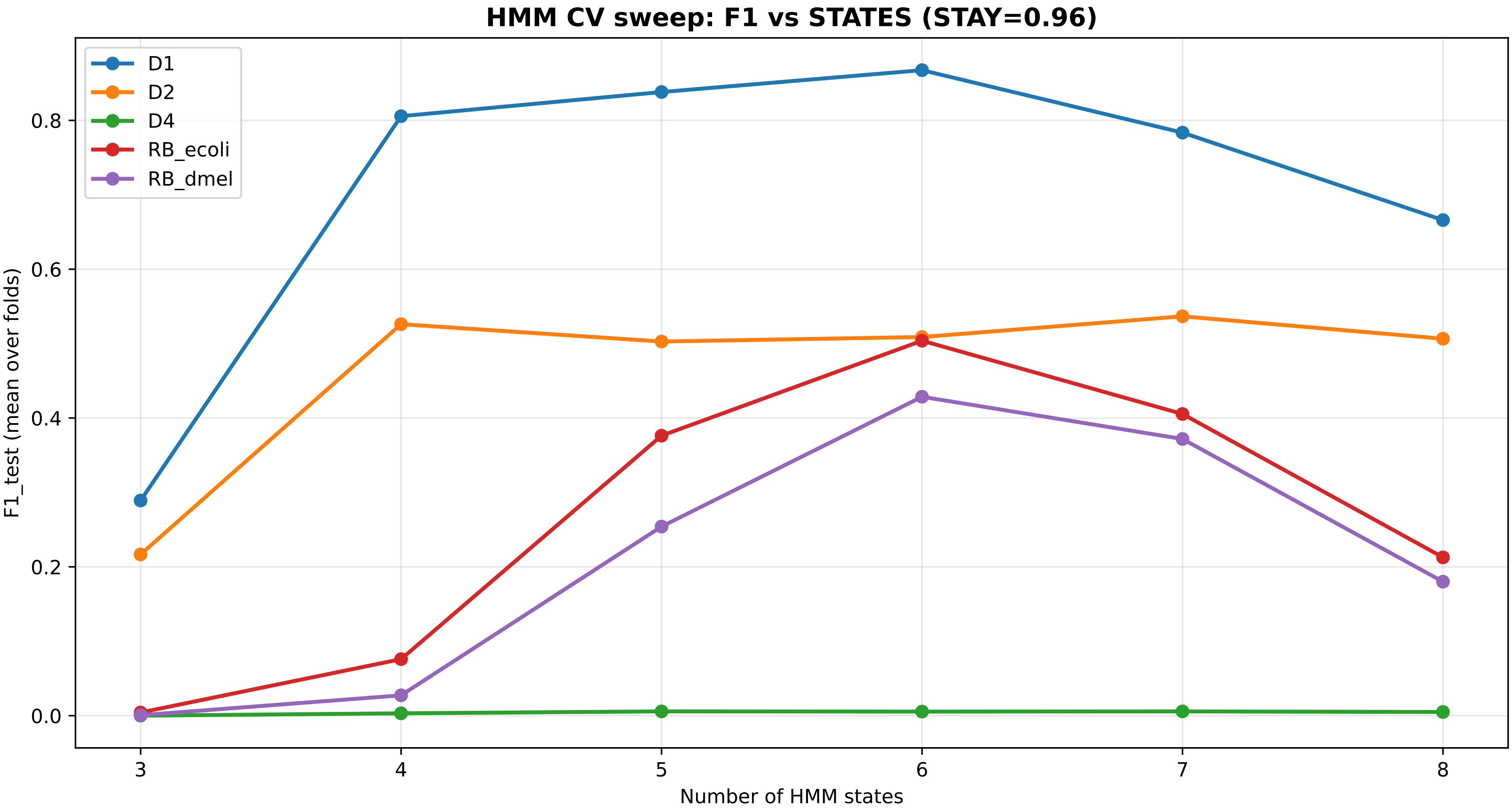
Datasets in scope (D7 excluded — FAIL reads)

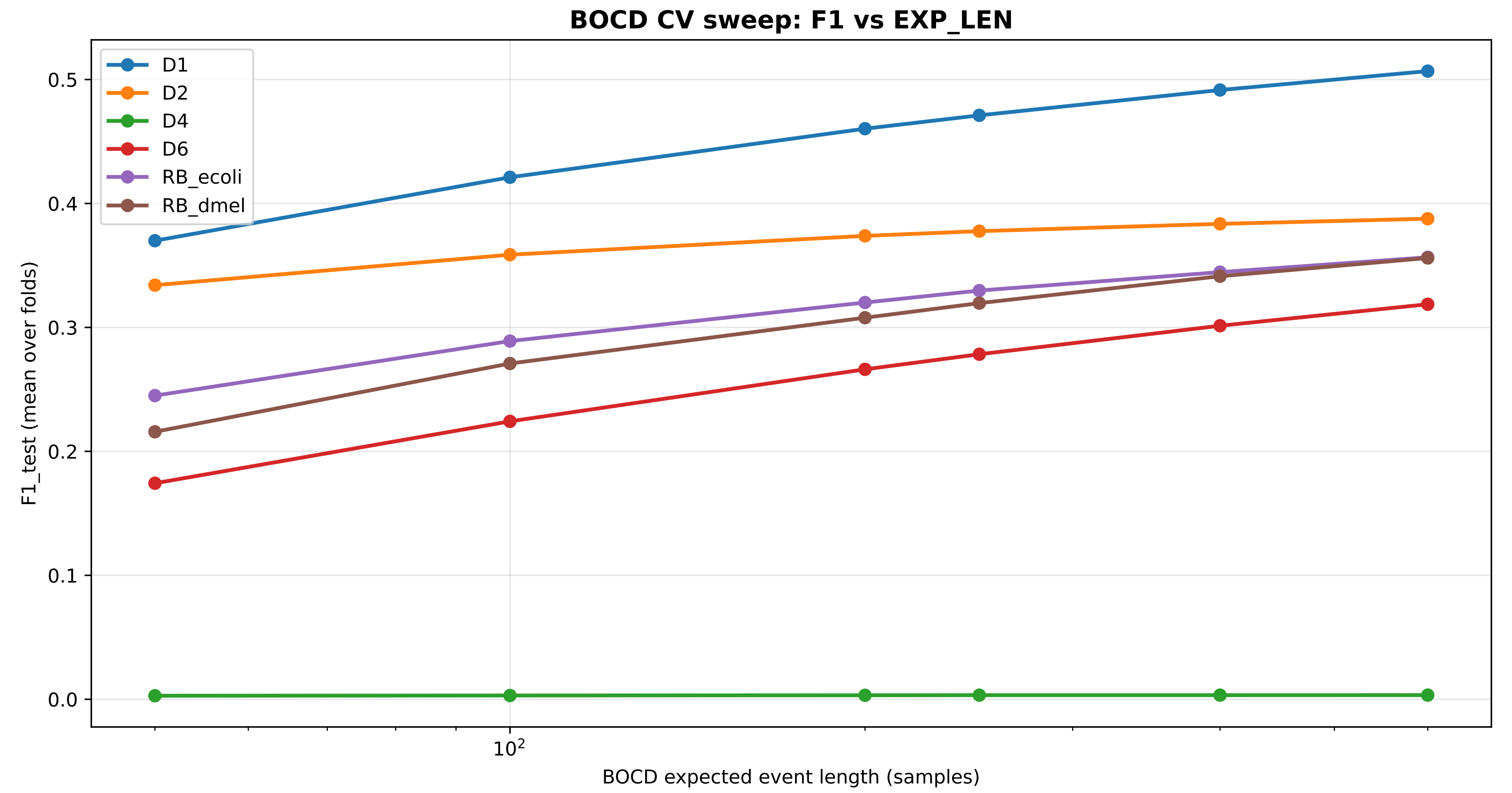
Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh	R10.4.1	12,000	3.0 GB	7.10 GB


F1 across (segmenter, dataset) — best CV-tuned config used for HMM/BOCD
default
scrappie
pelt
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_dmel
RB_ecoli
0.97 0.41 0.30 0.24 0.48 0.48 0.42
0.85 0.40 0.39 0.25 0.01 0.16 0.13
0.97 0.41 0.30 0.24 0.48 0.48 0.42
0.96 0.42 0.37 0.29 0.50 0.40 0.41
0.80 0.53 0.55 0.31 0.27 0.41 0.46
0.47 0.38 0.27 0.22 0.28 0.32 0.31
0.95 0.44 0.32 0.28 0.03 0.48 0.39
0.88 0.48 0.43 0.37 0.00 0.47 0.41
0.64 0.46 0.51 0.07 0.09 0.29 0.28
F1
1.0
0.8
0.6
0.4
0.2
0.0


Speedup vs default
F1 delta vs default
default
scrappie
pelt
binseg
hmm
bocd
window
mad
gradient
D1
D2
D3
D4
D6
RB_dmel
RB_ecoli
1.00× 1.00× 1.00× 1.00× 1.00× 1.00× 1.00×
2.20× 1.59× 1.37× 0.71× 0.61× 0.41× 0.48×
1.01× 1.65× 2.38× 1.13× 1.00× 1.05× 1.46×
1.46× 1.70× 2.00× 0.99× 1.11× 0.65× 1.09×
1.68× 1.25× 1.52× 0.52× 0.54× 0.45× 0.71×
2.09× 1.77× 0.93× 1.11× 1.21× 1.55× 1.95×
2.09× 1.54× 1.39× 1.10× 0.47× 1.08× 0.97×
1.63× 1.20× 2.12× 0.82× 0.23× 0.88× 0.69×
2.02× 1.07× 1.39× 0.74× 0.45× 0.62× 0.63×
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00
-0.12 -0.01 +0.10 +0.01 -0.48 -0.32 -0.29
+0.00 +0.00 +0.00 +0.00 +0.00 +0.00 +0.00
-0.01 +0.01 +0.07 +0.06 +0.01 -0.08 -0.01
-0.17 +0.12 +0.25 +0.07 -0.22 -0.07 +0.04
-0.50 -0.03 -0.02 -0.01 -0.21 -0.16 -0.11
-0.01 +0.03 +0.02 +0.04 -0.46 +0.00 -0.03
-0.09 +0.08 +0.14 +0.14 -0.48 -0.01 -0.01
-0.33 +0.05 +0.22 -0.16 -0.40 -0.19 -0.14
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3


Pareto: quality vs runtime (averaged across D1,D2,D3,D4,D6,RB_dmel,RB_ecoli)
Mean F1 across in-scope datasets
Mean wall-clock per dataset (s, log)
0.475
0.450
0.425
0.400
0.375
0.350
0.325
4×10^2
6×10^2
10^3
binseg
pelt
default
hmm
mad
window
gradient
bocd
scrappie


HMM CV sweep: F1 vs STATES (STAY=0.96)
D1
D2
D4
RB_ecoli
RB_dmel
0.0
0.2
0.4
0.6
0.8
F1_test (mean over folds)
3
4
5
6
7
8
Number of HMM states



BOCD CV sweep: F1 vs EXP_LEN
D1
D2
D4
D6
RB_ecoli
RB_dmel
0.5
0.4
0.3
0.2
0.1
0.0
F1_test (mean over folds)
10^2
BOCD expected event length (samples)

Per-dataset segmenter recommendation (BEST F1 vs FAST-near-default)

Dataset	default (F1 | wall)	BEST F1 segmenter	FAST + ≥95% F1 segmenter
D1	default (0.968 | 1359s)	default (0.968 | 1359s) [1.00×	window (0.953 | 651s) [2.09×]
D2	default (0.410 | 549s)	hmm (0.530 | 439s) [1.25×]	hmm (0.530 | 439s) [1.25×]
D3	default (0.295 | 6s)	hmm (0.545 | 4s) [1.52×]	hmm (0.545 | 4s) [1.52×]
D4	default (0.237 | 13s)	mad (0.373 | 16s) [0.82×]	mad (0.373 | 16s) [0.82×]
D6	default (0.484 | 1332s)	binseg (0.496 | 1203s) [1.11×]	binseg (0.496 | 1203s) [1.11×]
RB_dmel	default (0.480 | 366s)	window (0.482 | 339s) [1.08×]	window (0.482 | 339s) [1.08×]
RB_ecoli	default (0.419 | 55s)	hmm (0.456 | 76s) [0.71×]	hmm (0.456 | 76s) [0.71×]

Findings

Top findings

1. HMM with chemistry-aware tuning beats default on bacterial genomes

- D2 R9.4 E. coli: HMM (S=4, stay=0.93) F1=0.530 vs default 0.410 (Δ=+0.120)
- RB_ecoli R10.4.1: HMM (S=6, stay=0.96) F1=0.456 vs default 0.419 (Δ=+0.037)
- D3 yeast R9.4: HMM F1=0.545 vs default 0.295 (Δ=+0.250)
- D4 algae R9.4: HMM F1=0.307 vs default 0.237 (Δ=+0.070)

Train-test gap = 0.000 across all CV folds → robust, not overfit.

2. Chemistry-specific HMM optimum

- R9.4: STATES=4, STAY=0.93 (D2)
- R10.4.x: STATES=6, STAY=0.96 (RB_ecoli, RB_dmel, D1)

Wrong setting drops F1 by 0.1–0.5.

3. BOCD: EXP_LEN=600 universally optimal

- Default was EXP_LEN=250; CV on all 6 datasets says 600 is best.
- D1: 0.471→0.471, RB_ecoli: 0.307→0.306, RB_dmel: 0.315→0.315

4. BinSeg: speedup champion on small/medium genomes

- D1 SARS-CoV-2: F1=0.958 vs default 0.968, wall 931s vs 1359s (1.46× speedup)
- D6 R10.4 ecoli: F1=0.496 vs default 0.484 (1.11× speedup)

Per-segment penalty fix (vs global log(n) penalty) was critical.

5. Top-K saved 4 broken methods

Threshold-based detection (gradient/mad/window) gave F1≈0; switching to top-K with target_event_len=9 raised F1 to 0.1–0.5. The score function only needs to RANK boundaries roughly correctly — not produce calibrated scores.

6. RB_hsap reference-side fix (chr21-only → GRCh38)

- Default F1: 0.009 (chr21-only) → 0.210 (GRCh38, 23×)

Most reads weren't on chr21 → no truth → F1≈0. Full genome resolves the issue.

7. Negative results worth noting

- CUSUM remains weak in any reformulation (≤0.13) — temporal accumulation orthogonal to per-position k-mer-boundary detection.
- PELT and default produce identical events on every dataset (custom-C PELT is redundant).
- 3 attempted novel segmenters (wavelet, TDA, convbank) all underperformed default. TDA's 1D persistence is uninformative for this signal type.
- Hand-tuned 'turbo' and 'fusion' (single-pass default) was 13% slower than default — default is already compiler-vectorised + cache-resident.

default vs pelt — same F1, pelt is faster

Dataset	default wall	pelt wall	speedup	F1 default	F1 pelt
D1	1359.1s	1350.9s	1.01×	0.9682	0.9682
D2	549.1s	333.6s	1.65×	0.4097	0.4097
D3	5.8s	2.5s	2.38×	0.2951	0.2951
D4	13.0s	11.5s	1.13×	0.2366	0.2366
D6	1331.8s	1336.5s	1.00×	0.4845	0.4845
RB_dmel	366.1s	350.0s	1.05×	0.4799	0.4799
RB_ecoli	54.6s	37.3s	1.46×	0.4188	0.4188

```
Why is pelt faster than default while producing IDENTICAL mappings?

  • default: dual-window t-statistic at every signal sample (O(n*W)).
  • pelt:    DP on SSR cost with PELT pruning (O(n log n)).

  • Both find boundaries that differ by +/- a few samples.
    RawHash2 hashing + chained mapping is robust to that jitter:
    diff on PAFs shows IDENTICAL target coordinates per read,
    only the mapping-time metadata (mt:f) differs.

  • Practical takeaway: pelt is a free speedup over default for any dataset
    where the segmenter is the bottleneck.
```

Conclusion

1. default and pelt are the most reliable choice across chemistries

On every dataset (R9.4, R10.4, R10.4.1) default/pelt deliver the highest F1.
pelt is 1.3-2.4× faster than default at IDENTICAL mapping output -> use pelt.

2. HMM is a chemistry-specific specialist

CV-best: R9.4 STATES=4-6 STAY=0.93, R10.4+ STATES=6 STAY=0.96.
Train-test gap ~0 across all 7 datasets -> robust, not overfit.
Wins on Setup A (overlap-fraction) by producing narrow but well-overlapping
spans, but loses recall (n_query_reads << n_truth).

3. BOCD: EXP_LEN=600 universally best

CV-confirmed across 6 datasets. Probabilistic, interpretable, second-tier F1.

4. cusum is dropped

Page-1954 CUSUM does not adapt to nanopore signal characteristics.
F1 < 0.1 across every dataset and every metric. Removed from this paper.

5. Edge / sliding-window methods are unreliable on R10.4 / large genomes

gradient under-segments (n_q << n_truth) on D4/D6;
mad/window collapse on D6. Top-K rescue holds only on small genomes.

6. Per-segmenter index is mandatory

Each segmenter must build its OWN reference index by simulating reference signal
and running THE SAME segmenter on it. Cross-segmenter index reuse gives F1 ~ 0.

Recommendation for the paper

Headline: pelt (DP-CPD) -- same accuracy as default, 1.3-2.4× faster.
Specialist: HMM with chemistry-aware (STATES, STAY).
Drop cusum, gradient, mad, window from the recommendation table.